

CodeIgniter 3 Models for Municipal OTMS

This document contains recommended CI3 model structures for Municipal OTMS including Ward, Property, Owner, Tax, Payment, Mutation, and Reporting modules.

Project Structure

```
application/  
■■■ controllers/  
■■■ models/  
■■■ views/  
■■■ libraries/  
■■■ helpers/  
■■■ config/
```

Ward_model.php

```
class Ward_model extends CI_Model {  
    private $table = 'Ward';  
    public function get_all()  
    {  
        return $this->db->get($this->table)->result();  
    }  
    public function insert($data)  
    {  
        return $this->db->insert($this->table, $data);  
    }  
}
```

Property_model.php

```
class Property_model extends CI_Model {  
    private $table = 'Property';  
    public function get_all()  
    {  
        $this->db->select('Property.*, Ward.ward_name');  
        $this->db->from('Property');  
        $this->db->join('Ward', 'Ward.ward_id = Property.ward_id');  
        return $this->db->get()->result();  
    }  
    public function insert($data)  
    {  
        $this->db->insert($this->table, $data);  
        return $this->db->insert_id();  
    }  
}
```

Owner_model.php

```
class Owner_model extends CI_Model {
```

```
private $table = 'Owner';

public function insert($data)
{
    $this->db->insert($this->table, $data);

    return $this->db->insert_id();
}
}
```

Tax_model.php

```
class Tax_model extends CI_Model {

    private $table = 'Tax_Assessment';

    public function generate_tax($data)
    {
        return $this->db->insert($this->table, $data);
    }

    public function ward_wise_demand()
    {
        $this->db->select('
            Ward.ward_name,
            SUM(Tax_Assessment.total_tax) as demand
        ');

        $this->db->from('Tax_Assessment');

        $this->db->join(
            'Property',
            'Property.property_id = Tax_Assessment.property_id'
        );

        $this->db->join(
            'Ward',
            'Ward.ward_id = Property.ward_id'
        );

        $this->db->group_by('Ward.ward_id');

        return $this->db->get()->result();
    }
}
```

Payment_model.php

```
class Payment_model extends CI_Model {

    private $table = 'Tax_Payment';

    public function make_payment($data)
    {
        return $this->db->insert($this->table, $data);
    }

    public function collection_by_date($from, $to)
    {
        $this->db->select('
            payment_date,
            SUM(paid_amount) as total_collection
        ');

        $this->db->from($this->table);

        $this->db->where('payment_date >=', $from);
        $this->db->where('payment_date <=', $to);

        $this->db->group_by('payment_date');
        return $this->db->get()->result();
    }
}
```

```
}  
}
```

Mutation_model.php

```
class Mutation_model extends CI_Model {  
    private $table = 'Mutation';  
  
    public function apply_mutation($data)  
    {  
        return $this->db->insert($this->table, $data);  
    }  
  
    public function approve_mutation($mutation_id)  
    {  
        return $this->db  
            ->where('mutation_id', $mutation_id)  
            ->update($this->table, [  
                'approval_status' => 'APPROVED'  
            ]);  
    }  
}
```

Report_model.php

```
class Report_model extends CI_Model {  
  
    public function demand_collection_report($from, $to)  
    {  
        $this->db->select(''  
            Ward.ward_name,  
            SUM(Tax_Assessment.total_tax) as demand,  
            SUM(Tax_Payment.paid_amount) as collection  
        '' );  
  
        $this->db->from('Tax_Assessment');  
  
        $this->db->join(  
            'Property',  
            'Property.property_id = Tax_Assessment.property_id'  
        );  
  
        $this->db->join(  
            'Ward',  
            'Ward.ward_id = Property.ward_id'  
        );  
  
        $this->db->join(  
            'Tax_Payment',  
            'Tax_Payment.assessment_id = Tax_Assessment.assessment_id',  
            'left'  
        );  
  
        $this->db->group_by('Ward.ward_id');  
  
        return $this->db->get()->result();  
    }  
}
```

Example Controller

```
class Property extends CI_Controller {  
    public function __construct()  
    {
```

```
        parent::__construct();
        $this->load->model('Property_model');
    }
    public function index()
    {
        $data['properties'] =
            $this->Property_model->get_all();

        $this->load->view(
            'property/index',
            $data
        );
    }
}
```
